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Do markets overreact: International evidence [☆]

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Abstract

In this paper, using the Conrad and Kaul's methodology we test for the overreaction hypothesis – which maintains that stock prices systematically overshoot and therefore their reversal can be predicted from past performance – in seven industrialized countries. Consistent with the findings of Conrad and Kaul, we see no evidence of overreaction in the US. However, returns to long-term contrarian strategies in other countries seem to be generally significant. Moreover, we find that in the majority of the countries, while returns to arbitrage portfolios based on price are higher than those based on size, the latter generally outperform the winner–loser arbitrage portfolios. © 1999 Elsevier Science B.V. All rights reserved.

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1. Introduction

A large body of recent research has concluded that observed anomalous movements in stock prices, particularly the long-term reversals of extreme past stock price changes, can be explained by the corrections of initial overreactions to new information. This “stock market overreaction” hypothesis maintains that a given stock decreases (increases) too far in price because of recent bad (good) news associated with the stock, but eventually returns to its fundamental value as investors realize that they had overreacted (Lehmann, 1990; Shefrin and Statman, 1985). Hence, stocks that perform poorly (“losers”) relative to the market over a given period of time, referred to as the “portfolio formation period”, consistently outperform the market in the subsequent months or years, generally known as the portfolio “test” or “evaluation period”. Similarly, stocks with high returns (“winners”) during the portfolio formation period typically underperform the market during the evaluation period. Clearly, any evidence in favor of the overreaction hypothesis may indicate the presence of weak-form market inefficiency, since past stock price behavior would have predictive value favorable to contrarian strategies.

Substantial evidence consistent with the overreaction hypothesis was first provided by De Bondt and Thaler (1985). Observing that the changes in dividends alone are not large enough to explain the observed excess volatility in stock prices and that stock prices are highly correlated with changes in the following year’s earnings, they maintain that “investors seem to attach disproportionate importance to short-run economic developments”. They also cite the *P/E* ratio anomaly (i.e., that stocks with very low *P/E* ratios earn larger risk-adjusted returns than stocks with high *P/E* ratios) as a potential indicator of the existence of investor overreaction. Empirically, they show that the arbitrage portfolio earns 10.5% over one month subsequent to the portfolio formation date. The cumulative return to this portfolio over 36 months on the average is a whopping 24.6%.

Findings of a number of later studies have supported the overreaction hypothesis and both the short-term contrarian strategy (for example, see Lehmann, 1990; Jegadeesh, 1990) and the long-term contrarian strategy (Jegadeesh, 1990; De Bondt and Thaler, 1987; De Bondt and Thaler, 1990). A few studies, however, have either reported evidence that security analysts’ forecasts underreact to recent earnings (see Abarbanell and Bernard, 1992) or asserted that Value Line analysts underreact to the most recent earnings announcements (Mendenhall, 1991) or argued that market overreaction suggested by earlier research was actually a result of biases in computed returns (Conrad and Kaul, 1993). Although the overreaction hypothesis has occupied a central place in most studies, other possible explanations of stock market behavior have also been advanced. For example, the effects of the seasonality

in the returns to loser and winner portfolios (Chopra et al., 1992), of differences in risk levels (Chan, 1988; Ball and Kothari, 1989), and of the size effect (De Bondt and Thaler, 1987; Zarowin, 1990) have all been analyzed, though these issues are far from having been resolved.

Except for a few minor modifications, most studies use the methodology employed by De Bondt and Thaler (1985). However, in a recent paper Conrad and Kaul (1993) have argued that returns to the contrarian strategy employed in these studies are upwardly biased since they are obtained by cumulating single-period returns over long intervals. They replace the average cumulated abnormal returns with an alternative performance evaluation measure, the average holding period abnormal returns, and find that returns to the arbitrage portfolio are entirely due to the “January effect”, and not to overreaction. In fact, they show that in non-January months the arbitrage portfolio of losers and winners earns *negative* returns.

The purpose of this paper is to evaluate the performance of arbitrage portfolios based on past performance, price, and size in seven industrialized countries. Although the overreaction hypothesis has been tested in most of these countries, we for the first time apply the same methodology to all seven markets in a single study. We use holding period returns (HPR) both to sort stocks into winner and loser portfolios and to measure their test-period performance. We also attempt to determine the impact of price and market value (i.e., size) on HPR by employing Conrad and Kaul’s pooled cross-section-times series (CS-TS) approach. Moreover, we address Loughran and Ritter’s (Loughran and Ritter, 1996) recent criticism that CS-TS approach may confound cross-section patterns and time-series mean reversion patterns.

We find evidence in favor of long-term overreaction (i.e., significant positive return over a period of three years to the arbitrage portfolio based on past performance) in all countries, with the notable exception of the US. Further analysis of the data shows that in most countries, including Japan and UK, returns to winner and loser portfolios are closely related to price and not to size. Moreover, in these countries price-based arbitrage portfolios outperform those based on size or past performance. Therefore, in these countries overreaction might be a price-based phenomenon. We also find that returns to losers and/or winners are related to size or both to size and price in a few other countries. For example, in Italy arbitrage portfolio based on size earns a return greater than those based on price or past performance.

2. Data

All data used in this study are obtained from Worldscope Disclosure Database. They consist of five-year average and annual returns to a sample of

stocks in each country from 1982 to 1991, and are summarized in Table 1.¹ Consider the data for the US. The “annual return” indicates that we have the annual returns to 954 NYSE firms in 1982, 975 firms in 1983 and so on.² The “five-year return” shows the number of firms for which an average return for the past five years was available. Note that no data was available for the 1978–1982 period; we had average returns to 126 firms between 1979–1983, 916 firms between 1980–1984 and so on. It is clear that due to the lack of uniformity in the data across countries, the portfolio formation periods, the number of stocks in these periods as well as the number of test periods somewhat vary from one country to the next. For example, the five-year return data become available in 1986 for Italy, compared to 1983 for the US and 1984 for Canada or there are 1079 firms with five-year returns in the US in 1990, but only 65 in Italy, and so on.

3. Methodology

The statistical analysis used in this paper to test for overreaction in stock markets can be best described as a two-step procedure. In the first step, at the beginning of the test period, the winner and loser stocks are determined by their absolute performance over a five-year portfolio formation period, e.g., 1979–1983 in the US or 1980–1984 in Canada, and so on. This is accomplished simply by ranking the firms in terms of their performance as indicated by the five-year HPR data. Hence, in the US on the first portfolio formation date t (1983), top 35 firms are assigned to the winner portfolio, while the bottom 35 stocks make up the loser portfolio.³ This step is repeated for all subsequent overlapping five periods in each country. There are then eight portfolio formation periods in the US, beginning with 1979–1983 and ending with 1986–1990. Notice that in each case both winner and loser portfolios are formed conditional on past returns rather than price, earnings or other firm-generated

¹ In all countries, including the US, if the annual return to a company is given for a 12-month period not ending on 31 December, then the year assigned to this data is that one over which the majority of the data is generated. For example, if the annual return is given for the period 1 May 1990 to 30 April 1991, this observation is assigned to the calendar year 1990. Note that our findings for the US remained virtually unchanged when we reimplemented all empirical work with data that excluded all firms which do not have annual returns coinciding with calendar years.

² Note that the use of annual data precludes us from testing whether market overreaction, if present at all, is essentially a short-term phenomenon, i.e., whether “January” returns are significantly different from “non-January” returns.

³ Of course, as Table 1 shows, in some years the winner and loser portfolios consist of less than 35 stocks in Italy and Germany.

Table 1
Number of companies with five-year and annual returns by country

	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991
<i>US</i>										
Five-year return	0	126	916	942	962	982	1013	1042	1079	1127
Annual return	954	975	1006	1040	1073	1133	1212	1248	1248	1238
<i>Canada</i>										
Five-year return	0	0	100	160	162	163	175	196	211	226
Annual return	167	176	200	239	251	284	325	318	318	297
<i>UK</i>										
Five-year return	0	0	166	242	252	270	303	355	392	480
Annual return	258	278	327	403	451	644	1031	1181	1219	1066
<i>Japan</i>										
Five-year return	0	104	331	381	429	544	664	848	925	911
Annual return	457	549	661	846	932	1044	1133	1144	1148	966
<i>Germany</i>										
Five-year return	0	0	64	95	95	106	119	129	138	134
Annual return	96	107	121	133	140	155	245	275	286	259
<i>France</i>										
Five-year return	0	0	90	128	135	139	149	146	157	149
Annual return	137	148	158	159	175	182	360	387	393	354
<i>Italy</i>										
Five-year return	0	0	0	0	42	47	52	57	65	74
Annual return	44	50	57	60	68	91	167	185	190	168

The number of companies for which five-year and annual returns were available for each year between 1982–1991 in each country. The “five-year return” is the average return to the stock over the past five years. The US data include a sample of NYSE firms. That data were obtained from Worldscope Disclosure Database.

information. This method of ranking is widely employed in the literature (see De Bondt and Thaler (1985) and Conrad and Kaul (1993) among others).

The second step involves measuring the performance of winner and loser portfolios in each test period. The procedure employed by De Bondt and Thaler (1985) and others (for example, Kato, 1990) first determines the abnormal return (i.e., the return relative to the aggregate market) to each stock i in the winner (loser) portfolio in the first, second, and up to m -months or years in the portfolio evaluation period. This step is repeated for all subsequent test periods. Cumulated abnormal returns (CARs) are then obtained by adding up these abnormal returns, which in turn are used to calculate the average cumulated abnormal returns (ACARs) in each k , where $k = 1, \dots, m$, during the test period. Therefore, $ACAR_W(k)$ ($ACAR_L(k)$) indicates how much cumulated abnormal returns stocks in the winner (loser) portfolio earn on the average during k -months or years in the test period. The overreaction hypothesis predicts $ACAR_W(k) < 0$, $ACAR_L(k) > 0$, and most importantly, $ACAR_L(k) - ACAR_W(k) > 0$.

Conrad and Kaul (1993) have shown that this method is flawed in that it spuriously inflates the return to the arbitrage portfolio by cumulating short-term (monthly) returns to each stock over m months in both winner and loser portfolios. Since, as they argue, single-period returns are upwardly biased due to “measurement errors in observed prices due to bid–ask errors, non-synchronous trading, and/or price discreteness”, in cumulating monthly returns over long periods, one not only cumulates the true returns but also the upward bias in each monthly return.

Following Conrad and Kaul (1993), we use holding period returns (HPRs) in place of cumulated abnormal returns (CARs) as performance evaluation measure for all holding periods k , where $k = 1, 2$, and 3 years.⁴ The HPR of a winner (loser) stock over one year is $HPR(1) = R_1$, over two-year interval $HPR(2) = (1 + R_1)(1 + R_2) - 1$, and over three years $HPR(3) = (1 + R_1)(1 + R_2)(1 + R_3) - 1$. The average holding period abnormal returns (AHPARs) to the winner and loser portfolios then are:

$$\begin{aligned} AHPAR_W(k) &= \frac{1}{n} \sum_{i=1}^n HPR_{W_i}(k) - \frac{1}{N} \sum_{j=1}^N HPR_{M_j}(k) \\ &= AHPR_W(k) - AHPR_M(k), \end{aligned} \quad (1)$$

⁴ Obviously, the performance of the winner and loser portfolios can be evaluated only over one year, i.e., 1991, the last year in the data set, if the beginning of the test period is 1990 and over up to two years when the portfolios are formed in 1989. The test periods for all other portfolios extend up to three years.

$$\begin{aligned} \text{AHPAR}_L(k) &= \frac{1}{n} \sum_{i=1}^n \text{HPR}_{Li}(k) - \frac{1}{N} \sum_{j=1}^N \text{HPR}_{Mj}(k) \\ &= \text{AHPAR}_L(k) - \text{AHPAR}_M(k), \end{aligned} \quad (2)$$

$$i = 1, \dots, n \quad \text{and} \quad j = 1, \dots, N,$$

where $\text{AHPAR}_M(k)$ is the average holding period return to the market portfolio consisting of N stocks. Then the differences between the AHPARs of loser and winner portfolios are:

$$\text{DAHPAR}(k) = \text{AHPAR}_L(k) - \text{AHPAR}_W(k). \quad (3)$$

The overreaction hypothesis predicts $\text{AHPAR}_W(k) < 0$, $\text{AHPAR}_L(k) > 0$, and most importantly, $\text{DAHPAR}(k) > 0$. It can be shown that over all k -period returns, the bias in AHPARs will be the constant bias in a single period's return, compared to k times this amount in studies using CARs (see Conrad and Kaul, 1993).

As Loughran and Ritter's (1996) recent paper shows, however, the Conrad and Kaul methodology we use may suffer from a survivorship bias, since winner and loser portfolios include only those firms which survive into test periods. On the one hand, since low-priced stocks (often losers) are more likely to go bankrupt during the test period than high-priced stocks (usually winners), our procedure may nearly guarantee that low-priced stocks will have high returns during test periods. On the other hand, if losers tend to get taken over, the returns on the portfolio of losers may be understated, since in this case the takeover premium is ignored. In theory, then, it is not clear what kind of bias to the returns of the loser portfolio the survivorship bias will impart in a given country. Nevertheless, Loughran and Ritter (1996) show that the survivorship bias is indeed an important factor which overstates the importance of price variable in Conrad and Kaul's (1993) regressions for the US.

4. Empirical results

4.1. The overreaction hypothesis

Table 2 reports one-, two-, and three-year AHPARs of the loser and winner portfolios, and DAHPARs, the difference between the two. In all countries except the US and Canada, the loser portfolio consistently outperforms the market over one, two, and three years subsequent to the portfolio formation date, whereas the return to the winner portfolio is less than the return to the market portfolio in all three holding periods and in all countries. Most importantly, three-year DAHPARs are positive and statistically significant in all

Table 2
Average holding period abnormal returns of loser and winner portfolios of stocks, 1983–1991

Portfolio	1-year	2-year	3-year	Portfolio	1-year	2-year	3-year
<i>USA</i>				<i>Canada</i>			
Loser	-0.003 (0.036)	-0.037 (0.044)	0.022 (0.086)	Loser	-0.048* (0.013)	-0.025 (0.023)	-0.026 (0.013)
Winner	0.042 (0.053)	-0.001 (0.049)	-0.074* (0.016)	Winner	-0.003 (0.026)	-0.023 (0.026)	-0.151* (0.040)
DAHPARs	-0.045 (0.039)	-0.036 (0.070)	0.096 (0.097)	DAHPARs	-0.045 (0.036)	-0.002 (0.045)	0.124* (0.045)
<i>Japan</i>				<i>UK</i>			
Loser	0.024 (0.034)	0.136 (0.074)	0.259* (0.960)	Loser	0.085* (0.022)	0.199* (0.055)	0.331* (0.095)
Winner	-0.155* (0.062)	-0.366* (0.093)	-0.687* (0.119)	Winner	-0.032 (0.048)	-0.133 (0.101)	-0.254 (0.146)
DAHPARs	0.180** (0.093)	0.502* (0.163)	0.945* (0.200)	DAHPARs	0.118 (0.065)	0.333** (0.140)	0.585** (0.237)
<i>Germany</i>				<i>France</i>			
Loser	0.031 (0.044)	0.127 (0.113)	0.266** (0.101)	Loser	0.121* (0.046)	0.241 (0.137)	0.374 (0.191)
Winner	-0.047 (0.030)	-0.115 (0.100)	-0.238** (0.114)	Winner	-0.032 (0.019)	-0.139* (0.029)	-0.255* (0.020)
DAHPARs	0.078 (0.071)	0.242 (0.210)	0.505** (0.210)	DAHPARs	0.152* (0.047)	0.380* (0.115)	0.629* (0.182)

<i>Italy</i>			
Loser	0.007 (0.026)	0.050 (0.027)	0.093 (0.037)
Winner	–0.018 (0.011)	–0.048** (0.020)	–0.122* (0.023)
DAHPARs	0.024 (0.029)	0.098 (0.047)	0.216* (0.053)

AHPARs to the winner and loser portfolios of stocks over one, two, and three years subsequent to the portfolio formation date, and DAHPARs, i.e., the differences between the two. The numbers reported are averages of AHPARs of overlapping intervals. The numbers in parentheses are standard errors, computed using the Newey–West (1987) adjustment for heteroskedasticity and autocorrelation.

* Significant at the 5% level.

** Significant at the 10% level.

countries except the US. All these findings are consistent with the overreaction hypothesis. Since with few exceptions only three-year DAHPARs (and AHPARs) are statistically significant, the evidence seems to favor long-term contrarian strategies in all countries except the US. The average return to the arbitrage portfolio of losers and winners is 94.5% in Japan, 62.9% in France, 58.5% in UK, 50.5% in Germany, 21.6% in Italy, and 12.4% in Canada. The results obtained from the US data are generally consistent with the findings of Conrad and Kaul (1993). AHPARs to the winner and loser portfolios as well as DAHPARs often have the signs not predicted by the overreaction hypothesis or are not statistically significant.

There have been other studies that have attempted to seek evidence for the overreaction hypothesis in countries other than the US. But since sample periods, data sources, and most importantly, methodologies differ, it is difficult to compare our findings with these others. Nevertheless, it might be interesting to note that Kryzanowski and Zhang (1992) reported statistically significant continuation behavior in the short-term (the next one or two years into test periods) for winners and losers, but insignificant reversal behavior for both winners and losers over longer periods in Canada; Stock (1990) found evidence of long-term overreaction in Germany and Dissanaike (1997) in UK; and Kato (1990), using De Bondt and Thaler methodology, showed that in Japan most of the overreaction occurs in the winner portfolio, as we do.

A notable aspect of our findings is that the overreaction effect is, in some cases, strongly asymmetric. For example, the price reversal for winners is much larger in Japan than it is for losers. Specifically, over three years winner portfolios earn 68.7% less than the market, whereas loser portfolios of 35 stocks earn, on average, 25.9%. This asymmetric reaction is less pronounced in countries like UK, Germany and France, where losers outperform the market to a somewhat greater extent than winners underperform it. A few explanations have been advanced for the apparent asymmetry found in other studies as well. Initially, De Bondt and Thaler (1985), who found larger overreaction effect for losers in the US, had argued that differences in systematic risk could explain the asymmetry. Later, Conrad and Kaul (1993) tried to explain it in terms of bid–ask biases. Finally, Dissanaike (1996) maintained that the observed asymmetry might be entirely “illusory” and may be explained by the peculiar properties of returns, namely that “the test-period returns on a contrarian portfolio is not always a reliable measure of the strength of price reversals”. We think that one possible explanation for the strong asymmetry found in Japan might be that Japan had a fairly substantial market run-up through the 1980s, while European markets were relatively less bullish. If in boom markets investors tend to overreact more on the upside than on the downside – i.e., if any good news is great news and bad news is not so bad – then the winners would have been bid way up, while losers are not bid too far down.

4.2. Price- and size-based portfolios

In this section we first describe the characteristics of loser and winner companies in terms of their stock prices and market values, and then evaluate the performance of 35 low- and high-market value companies as well as 35 low- and high-price stocks without considering their past returns. Table 3 shows the price and market value estimates of loser and winner firms based on average values at each portfolio formation date, i.e., at the beginning of each evaluation period. No sophisticated statistical analysis is needed to observe that winner firms have higher average prices and market values than losers in all countries. For example, in the US the mean (median) price of loser firms is \$8.014 (\$7.390), compared to the mean (median) price of \$22.206 (\$21.946) for winners. Even in Italy, where the ratio of the mean prices of winners and losers is the smallest, the mean price for winners exceeds that of losers nearly by 850 lire. Similarly, while the mean (median) market value of winners in the US is \$1,674.278 million (\$1,737.601 million), for losers this figure is only \$197.979 million (\$116.475 million).

One interesting question, therefore, pertains to the relationship between the holding period returns to loser and winner stocks and their prices and market values (size). Another involves the performance of price- and size-based arbitrage portfolios. The remainder of this paper is devoted primarily to these issues. To begin with, we estimate pooled CS-TS regressions of three-year HPR of individual stocks in each of the loser as well as winner portfolios on their prices and market values at the beginning of each evaluation period. The results are reported in Table 4. Secondly, we evaluate the performance of low- and high-priced as well as low- and high-market value firms exactly the same way we evaluated the performance of winner and loser firms and summarize our findings in Table 5 and Table 6 below.

Table 4 shows the pooled CS-TS regressions of HPR on price (regressions (1)), on size (regressions (2)), and on both price and size (regressions (3)). We compute the *t*-statistics (given in parentheses) using the Newey–West (1987) adjustment for heteroskedasticity and autocorrelation, using a lag of 2. Now, Loughran and Ritter (1996) argue convincingly that such pooled CS-TS regressions may confound cross-sectional patterns with time-series mean reversion patterns. They reestimate regressions of HPR for the US data using Fama and MacBeth (1973) type approach, and obtain much smaller (and statistically insignificant) average coefficient on price. They therefore conclude that pooled CS-TS approach overstates the importance of price in explaining HPR, since in addition to picking up cross-sectional patterns, this approach “benefit from time-series mean reversion in prices as well”. We try to address this issue by estimating the regressions of HPR on both price and size using Fama–MacBeth approach and report our results in Table 4 (regressions (4)).

Table 3
The price and market value estimates based on average values at the beginning of each evaluation period

	Price (local currency)				Market value (millions of local currency)			
	Mean	Median	Minimum	Maximum	Mean	Median	Minimum	Maximum
<i>USA</i>								
Loser	8.014	7.390	2.096	17.147	197.979	116.475	26.09	759.427
Winner	22.20	21.94	11.635	33.939	1674.278	1737.601	326.672	4064.197
<i>Canada</i>								
Loser	9.462	10.792	3.643	12.030	386.599	443.323	86.420	560.394
Winner	21.788	20.159	15.831	31.497	1029.359	1058.484	869.213	1176.506
<i>Japan</i>								
Loser	1144.31	1046.98	339.60	2091.92	260.01	254.75	138.64	478.89
Winner	2416.59	2649.88	1194.60	3287.38	1178.42	829.71	302.37	2847.24
<i>UK</i>								
Loser	1.597	1.729	0.890	2.125	351.553	449.654	123.100	559.718
Winner	2.732	2.732	1.693	3.547	838.304	794.325	424.626	1387.713
<i>Germany</i>								
Loser	332.235	269.631	188.588	566.23	2309.113	1224.543	903.975	6318.551
Winner	632.713	662.593	277.471	873.031	5126.435	4243.184	1630.799	8760.300
<i>France</i>								
Loser	359.889	296.413	144.014	643.853	2239.639	2023.446	1006.136	3446.854
Winner	715.982	693.320	460.882	1214.540	5964.850	5695.653	2892.840	9351.120
<i>Italy</i>								
Loser	8884.85	8827.31	5969.66	11118.49	1380.82	1248.02	956.76	2306.51
Winner	9743.40	10195.32	7992.56	12062.12	3533.14	3682.22	2656.38	4028.86

Table 4
Regressions of holding period returns on price and market values of loser and winner stocks, 1983–1992

	$\hat{\alpha}_i$	$\hat{\beta}_i$	$\hat{\delta}_i$	$\bar{R}^2$	Winner	$\hat{\alpha}_i$	$\hat{\beta}_i$	$\hat{\delta}_i$	$\bar{R}^2$
<i>USA</i>									
Loser	(1)	0.534* (3.03)	-0.296* (-4.16)	0.056	Winner	(1)	1.124* (4.57)	-0.456* (-5.47)	0.164
	(2)	0.586 (0.85)	-0.050 (-0.85)	-0.001		(2)	-0.407 (-1.02)	0.025 (0.83)	-0.003
	(3)	-0.137 (-0.20)	-0.352* (-4.27)	0.068 (1.06)		(3)	-0.519 (-1.29)	-0.582* (-4.73)	0.218
	(4)	0.298 (0.312)	-0.412* (-2.63)	0.034 (0.33)		(4)	-0.301 (-1.30)	-0.709* (-7.72)	0.260
<i>Canada</i>									
Loser	(1)	0.528** (1.86)	-0.261* (-2.26)	0.057	Winner	(1)	0.506* (2.98)	-0.237* (-4.09)	0.143
	(2)	0.771 (0.88)	-0.065 (-0.94)	0.003		(2)	-0.146 (-0.39)	-0.001 (-0.02)	-0.007
	(3)	0.393 (0.47)	-0.270* (-2.27)	0.012 (0.177)		(3)	0.028 (0.078)	-0.253* (-4.03)	0.146
	(4)	0.075 (0.09)	-0.248** (-2.25)	0.032 (0.40)		(4)	0.139 (0.76)	-0.329* (-3.79)	0.152
<i>Japan</i>									
Loser	(1)	3.069* (4.54)	-0.439* (-4.46)	0.120	Winner	(1)	1.127* (2.20)	-0.243* (-3.51)	0.060
	(2)	3.196* (2.35)	-0.157* (-2.18)	0.016		(2)	0.507 (0.58)	-0.059 (-1.36)	0.007
	(3)	3.133* (2.33)	-0.437* (-3.74)	0.115 (0.046)		(3)	1.465 (1.76)	-0.231* (-2.95)	0.056
	(4)	4.784* (2.74)	-0.602* (-2.35)	0.176 (-0.39)		(4)	0.560 (0.55)	-0.295* (2.50)	0.194

Table 4 (Continued)

	$\hat{\alpha}_i$	$\hat{\beta}_i$	$\hat{\delta}_i$	$\bar{R}^2$	Winner	$\hat{\alpha}_i$	$\hat{\beta}_i$	$\hat{\delta}_i$	$\bar{R}^2$
<i>UK</i>									
Loser									
(1)	0.421* (4.24)	-0.371* (-2.57)		0.096	Winner	-0.147* (-2.06)	-0.168* (-3.12)		0.032
(2)	2.504* (3.22)		-0.182* (-2.95)	0.076		-0.055 (-0.11)		-0.017 (-0.43)	-0.005
(3)	1.413** (1.89)	-0.270* (-2.24)	-0.085 (-1.36)	0.100		-0.224 (-0.45)	-0.171* (-3.14)	0.006 (0.16)	0.026
(4)	1.216 (1.33)	-0.135 (-1.01)	-0.081 (-0.95)	0.048		-0.119 (-0.23)	-0.307* (-3.49)	0.006 (0.15)	0.110
<i>Germany</i>									
Loser									
(1)	2.433* (4.83)	-0.402* (-4.69)		0.073	Winner	0.975* (2.98)	-0.203* (3.89)		0.060
(2)	2.362* (2.77)		-0.156* (-2.55)	0.036		1.713* (2.72)		-0.133* (-3.29)	0.092
(3)	3.331* (3.76)	-0.346* (-4.50)	-0.089 (-1.59)	0.082		2.107* (3.21)	-0.141* (-2.37)	-0.102* (-2.13)	0.101
(4)	2.777* (2.20)	-0.307** (-2.62)	-0.065 (1.36)	0.044		2.261* (4.77)	-0.141* (-6.69)	-0.113* (-3.28)	0.111
<i>France</i>									
Loser									
(1)	1.717** (1.91)	-0.268 (-1.69)		0.018	Winner	0.628 (1.48)	-0.148* (2.24)		0.035
(2)	4.443* (3.03)		-0.302* (-3.01)	0.054		1.528** (1.73)		-0.120* (2.15)	0.028
(3)	4.481* (2.97)	-0.089 (-0.57)	-0.272* (-2.81)	0.049		2.333* (2.30)	-0.144* (-2.18)	-0.116* (-2.11)	0.061
(4)	3.996* (2.22)	-0.052 (-0.34)	-0.247* (-2.84)	0.042		3.258 (1.90)	-0.166** (-2.26)	-0.173 (-1.85)	0.108

Italy Loser	(1)	0.837* (2.21)	–0.088* (–2.14)	0.029	Winner	(1)	0.518 (–1.11)	–0.074 (–1.37)	0.005
	(2)	–0.348 (–0.25)	0.022 (0.32)	–0.017		(2)	2.806* (3.67)	–0.138* (–3.97)	0.152
	(3)	–0.533 (–0.43)	–0.110* (–2.44)	0.030		(3)	2.793* (3.57)	0.005 (0.07)	0.136
	(4)	–0.984 (–1.07)	–0.104 (–1.16)	–0.007		(4)	2.731* (3.90)	–0.021 (–0.05)	0.111

Regressions of three-year holding period returns of loser and winner stocks measured over three-year overlapping periods on their prices and market values at the beginning of each three-year interval. The estimated regression is

$$\text{HPR}_{3,yt} = \alpha_t + \beta_1 P_{it} + \delta_1 M_{it} + \epsilon_{3,yt},$$

where HPR is the three-year holding period return of losers and winners, P and M are the logarithms of beginning of the period price and market value of the losers and winners, and ϵ is a random error term. For each country, and for both loser and winner portfolios, pooled CS-TS regressions estimates are given in rows (1), (2) and (3). In rows (4), the results of Fama–MacBeth approach, where coefficients reported are the averages of cross-sectional regressions, are given. In all regressions, numbers in parentheses are t -statistics, computed using the Newey–West (1987) procedure, with a lag of 2.

* Significant at the 5% level.

** Significant at the 10% level.

Table 5
Average holding period abnormal returns of low- and high-priced portfolios, 1983–1991

Portfolio	1-year	2-year	3-year	Portfolio	1-year	2-year	3-year
<i>USA</i>				<i>Canada</i>			
Low-price	0.194* (0.075)	0.570* (0.110)	1.145* (0.140)	Low-price	0.177* (0.027)	0.412* (0.081)	0.712* (0.163)
High-price	−0.157* (0.020)	−0.286* (0.034)	−0.433* (0.065)	High-price	−0.099* (0.027)	−0.211* (0.059)	−0.347* (0.119)
DAHPARs	0.351* (0.079)	0.856* (0.119)	1.578* (0.191)	DAHPARs	0.276* (0.051)	0.622* (0.138)	1.059* (0.280)
<i>Japan</i>				<i>UK</i>			
Low-price	0.273* (0.108)	0.621** (0.220)	1.065* (0.386)	Low-price	0.317* (0.041)	0.680* (0.165)	0.923* (0.277)
High-price	−0.156** (0.068)	−0.349* (0.139)	−0.620* (0.211)	High-price	−0.143* (0.050)	−0.329* (0.122)	−0.552** (0.232)
DAHPARs	0.429* (0.157)	0.970* (0.383)	1.685* (0.532)	DAHPARs	0.460* (0.084)	1.009* (0.272)	1.474* (0.474)
<i>Germany</i>				<i>France</i>			
Low-price	0.083* (0.032)	0.207** (0.104)	0.292* (0.104)	Low-price	0.181* (0.027)	0.520* (0.125)	0.996* (0.236)
High-price	−0.024 (0.028)	−0.042** (0.086)	−0.083 (0.133)	High-price	−0.146* (0.028)	−0.398* (0.101)	−0.684* (0.189)
DAHPARs	0.106** (0.056)	0.249 (0.184)	0.374 (0.232)	DAHPARs	0.327* (0.045)	0.918* (0.223)	1.680* (0.422)

<i>Italy</i>			
Low-price	–0.001 (0.025)	0.058 (0.057)	0.134 (0.075)
High-price	–0.094* (0.024)	–0.190* (0.048)	–0.250* (0.088)
DAHPARs	0.094** (0.045)	0.247* (0.098)	0.384* (0.163)

AHPARs to the low- and high-priced portfolios of stocks, and DAHPARs, i.e., the differences between the two. The numbers reported are averages of AHPARs of overlapping intervals. The numbers in parentheses are standard errors, computed using the Newey–West (1987) adjustment for heteroskedasticity and autocorrelation.

* Significant at the 5% level.

** Significant at the 10% level.

Table 6
Average holding period abnormal returns of sized based portfolios, 1983–1991

Portfolio	1-year	2-year	3-year	Portfolio	1-year	2-year	3-year
<i>USA</i>				<i>Canada</i>			
Low-MVE	0.063 (0.050)	0.077 (0.042)	0.136* (0.022)	Low-MVE	0.148* (0.025)	0.242* (0.056)	0.379* (0.122)
High-MVE	0.023 (0.029)	0.097* (0.036)	0.190* (0.055)	High-MVE	−0.060** (0.032)	−0.131 (0.073)	0.204 (0.143)
DAHPARs	0.040 (0.075)	−0.021 (0.063)	−0.054 (0.061)	DAHPARs	0.208* (0.038)	0.372* (0.086)	0.583** (0.241)
<i>Japan</i>				<i>UK</i>			
Low-MVE	0.226* (0.085)	0.562* (0.173)	0.841* (0.236)	Low-MVE	0.204* (0.044)	0.496 (0.176)	0.803 (0.436)
High-MVE	−0.089 (0.066)	−0.206 (0.131)	−0.365** (0.178)	High-MVE	−0.013 (0.064)	−0.064 (0.150)	−0.113 (0.247)
DAHPARs	0.315* (0.120)	0.769* (0.249)	1.205* (0.381)	DAHPARs	0.217* (0.086)	0.560 (0.312)	0.916 (0.677)
<i>Germany</i>				<i>France</i>			
Low-MVE	0.035 (0.021)	0.070 (0.044)	0.171* (0.045)	Low-MVE	0.135* (0.028)	0.377* (0.101)	0.746* (0.197)
High-MVE	−0.030 (0.044)	−0.098 (0.102)	−0.171 (0.146)	High-MVE	−0.025 (0.029)	−0.176* (0.074)	−0.377* (0.145)
DAHPARs	0.065 (0.056)	0.168 (0.143)	0.342 (0.189)	DAHPARs	0.160* (0.054)	0.553* (0.173)	1.123* (0.339)

<i>Italy</i>			
Low-MVE	0.082* (0.024)	0.197* (0.046)	0.304* (0.098)
High-MVE	–0.064* (0.024)	–0.161* (0.050)	–0.201* (0.069)
DAHPARs	0.146* (0.042)	0.359* (0.083)	0.505* (0.154)

AHPARs to the low- and high-MVE portfolios of firms, and DAHPARs, i.e., the differences between the two. The numbers reported are averages of AHPARs of overlapping intervals. The numbers in parentheses are standard errors, computed using the Newey–West (1987) adjustment for heteroskedasticity and autocorrelation.

* Significant at the 5% level.

** Significant at the 10% level.

First consider the pooled CS-TS regressions. For both loser and winner portfolios, in the US, Canada, and Japan, the coefficient of the price variable is statistically significant in regressions (1), whereas in regressions (2) the size coefficient is insignificant. Also the $\bar{R}^2$ estimates for the price regressions are much larger than those for the market value estimates. In regressions (3) of HPR on both price and size, $\hat{\beta}$'s are again statistically significant with negative signs. Hence, in these countries holding period returns to all stocks, regardless of their past performance, increase as price declines. One possible implication of these findings is that, in countries like Canada and Japan, where we found evidence that the returns to the loser–winner arbitrage portfolio are significantly positive and where there is strong relation between HPR and past prices, overreaction might be primarily a price (especially low-price) phenomenon.⁵ In the UK, too, all these results hold. The only exception is that the market value coefficient for loser firms is significant in simple regression (2), though it is rendered insignificant in regression (3). Again, overreaction essentially seems to be a price-based phenomenon. Also note that in the US, where we found no evidence of overreaction, HPR to loser firms are negatively related to price, whereas HPR to winners are negatively related to price and positively related to size.

In the continental European countries in our sample, the results are somewhat different and much less clear. For example, in Italy $\hat{\beta}$'s are statistically significant only for loser firms, again with negative signs, but for winners HPR are related to size alone. Specifically, the larger the size of the winner firm, the smaller the HPR. In France and Germany, HPR to winners are related negatively to size, as is Italy. But in France a stronger negative relation between HPR to losers and size suggests that overreaction might be to some extent a small-firm phenomenon. Finally, in Germany while $\hat{\beta}$'s are significant in regressions (3) for losers, both $\hat{\beta}$'s and $\hat{\delta}$'s are significant for winners, implying both price and size effects for the latter.

Now compare the coefficient estimates in the pooled CS-TS regressions (3) and the Fama–MacBeth type regressions (4) in Table 4. Notice that in general the estimates of market value coefficients are not as sensitive to the two alternative approaches as the price coefficient estimates are – a finding reported also by Loughran and Ritter (1996) in their study of the US markets. While in winner portfolios the importance of size in explaining HPR may somewhat decline (France and Italy), slightly rise (USA and Canada), or remain insignificant (Japan, UK, and Germany), the use of the

⁵ We are aware that strong associations found between price or size and actual performance, though suggesting, may not necessarily weaken the support found for overreaction. After all, these associations are without much basis in current theory and no one seems to understand well why they exist, either in a fully rational or in a noise trader framework.

Fama–MacBeth approach has very little impact on the significance of size in explaining returns to loser portfolios. However, in winner portfolios the estimates of the price coefficient are larger in most countries, including the US. In loser portfolios, the price may increase (USA, Japan) or decrease in importance (Canada, UK, Germany, Italy). At least as far as the US is concerned, these findings are surprising: while Loughran and Ritter (1996) find a pooled CS-TS approach to overstate the importance of price in explaining cross-sectional returns, we find the opposite result. We think that the differences in the sample periods used and in the specification of the regression equations in our studies, the possible distorting effect of survivorship bias in our regression, and the fact that we estimate separate regressions for winner and loser portfolios are all factors that might contribute to this discrepancy in our findings.

In order to gain further insight into the relationship between HPR and price and size, we next implement two simple strategies. First, the performances over three years of low- and high-priced stocks are evaluated in each country. Specifically, stocks are ranked according to their prices at the beginning of portfolio test periods – without considering their past performance – and the top 35 and bottom 35 firms are assigned to high- and low-price portfolios. Then the performance of these portfolios is evaluated using AHPARs for exactly the same three-year periods as the loser and winner portfolios. The results are shown in Table 5. Secondly, the same procedure is repeated for size-based portfolios, i.e., portfolios of 35 largest and 35 smallest firms at the time portfolios are formed. Table 6 reports the AHPARs to market value-based portfolios.

Table 5 shows that in all countries, except Italy, low-price (high-price) portfolios outperform (underperform) the market over one, two, and three years into the evaluation period – even in Italy, the arbitrage portfolio of low- and high-price stocks earns an average return of 38.4% over three years. In the US, for example, after three years the low-price portfolio outperforms the market by 144.5%, while the high-price portfolio earns an average abnormal return of –43.3%. Hence, the average return to the price-based arbitrage portfolio during the sample period is 157.5%. In fact, in all countries, except Germany, DAHPARs to price-based long-term strategies are larger than DAHPARs to those based both on past performance and market value. Moreover, again with the exception of Italy, in countries with statistically significant three-year DAHPARs, market value-based portfolios, although outperformed by price-based portfolios, still earn higher average abnormal returns than performance-based portfolios. For example, in Canada three-year DAHPAR to price-based portfolio is 105.9%, in Japan 168.5%, and in France 168%. These figures for market value- and performance-based portfolios are, respectively, 58.3% and 12.4% in Canada, 120.5% and 94.5% in Japan, and 112.3% and 62.9% in France.

To sum, we find evidence of overreaction in all countries except the US. These findings are in general consistent with the results reported in the overreaction literature. We also find that, with the exception of Germany, the price- and size-based arbitrage portfolios earn a larger return than loser–winner portfolios suggested by the overreaction hypothesis. For one thing, these results suggest that overreaction effects might be explained in part by price and size effects. Secondly, they should be viewed in light of Loughran and Ritter's (1996) conclusion that when portfolios are formed on a single variable such as past performance, price or size, "the combined effects of correlated variables that are related to returns are present, overstating the impact of the single variable". Of course, in some cases the contrarian strategy has been shown to earn insignificant abnormal returns after the time-varying beta risk for the winner and loser stocks are adjusted for (Chan, 1988; Ball and Kothari, 1989; Chen and Sauer, 1997).⁶ In all countries we assume that winners and losers are of equivalent risk, and do not address the "difficult question" of how much of the higher average returns to loser portfolios can be accounted for risk-bearing (see Loughran and Ritter, 1996; Jones, 1993).

5. Conclusion

In this paper, we first test the overreaction hypothesis in seven industrialized countries using the methodology suggested by Conrad and Kaul. We then try to determine whether price or size is significant in explaining holding period returns to loser and winner portfolios. Finally, we evaluate the performance of arbitrage portfolios based on price and size, and compare the findings to the performance of loser–winner arbitrage portfolios. While we find no evidence supporting the overreaction hypothesis in the US, we see that low-price portfolios consistently outperform the market, while high-price portfolios earn a return below the aggregate market. These findings are similar to those obtained by Conrad and Kaul (1993). In other countries we find evidence of overreaction, though this effect is relatively weak in Canada. Generally speaking, these results are also consistent with other evidence reported in the literature. However, since long-term investment strategies based on size and especially price produce returns higher than those based on past performance, and since losers (winners) tend to be low (high) price and low (high) market value firms, price and size effects might explain some of the long-term price reversals observed in winner and loser stocks. Moreover, two points must be considered in interpreting our findings: first, as Loughran and Ritter (1996) observed, when

⁶ For countries other than the US, the differential risk does not always seem to be a possible explanation for higher returns to losers. For example, see Dissanaike's (1997) UK study.

portfolios are formed on a single variable like past returns, price or size, the impact of any of these variables will probably be overstated. Secondly, the returns to some portfolios (i.e., losers) might be accounted, at least in part, by higher risk.

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